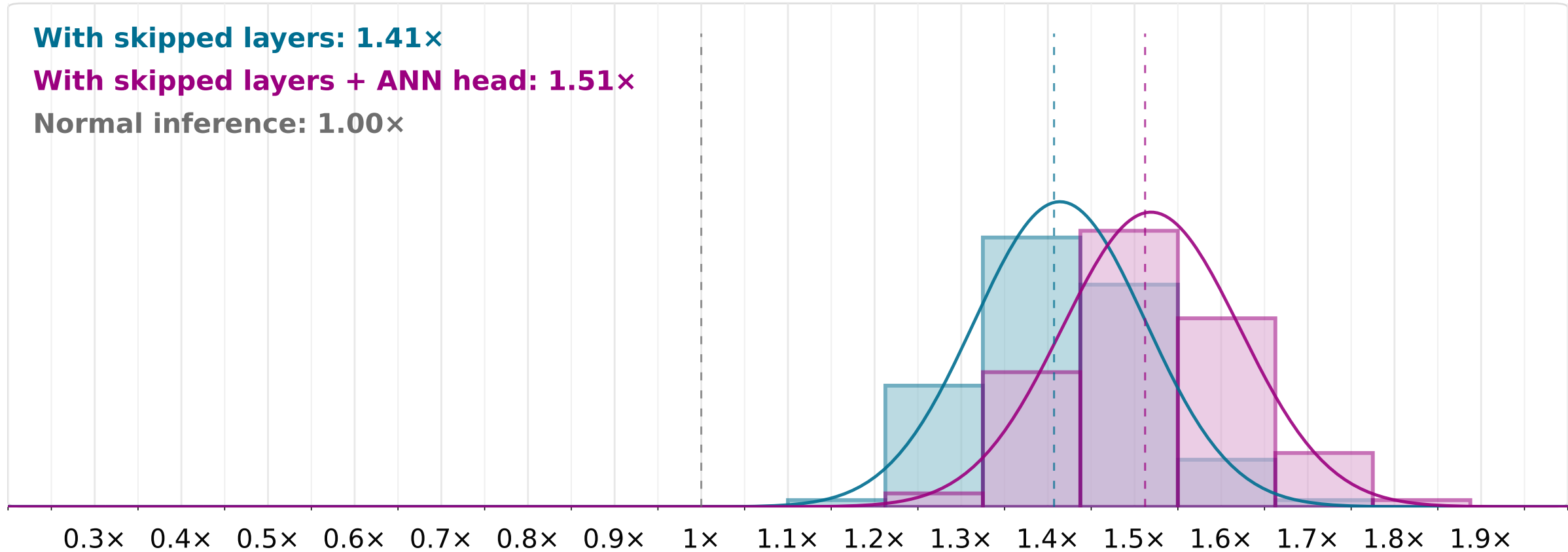


Self-speculation per-token speedup

Llama 3.1 8B Instruct



Setup

Prompt set = **python diverse**
Gap = **(1, 1)**
Block size = **2**
Layers = **32**
LM vocab = **128,256**
LM params = **8.03B**
Head params = **525.3M**
Head portion = **6.54%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **100**

Results

Peak mem normal = **15.06 GiB**
Mean speedup (w/o ANNH) = **1.41×**
Acceptance rate (w/o ANNH) = **44.3%**
Exact match (w/o ANNH) = **57.0%**
Peak mem self (w/o ANNH) = **15.07 GiB**
Mean speedup (w ANNH) = **1.51×**
Acceptance rate (w ANNH) = **43.7%**
Exact match (w ANNH) = **57.0%**
Peak mem self (w ANNH) = **15.14 GiB**
Tokens gen = **s 20,000 / n 20,000**

Profile

Drafter split (w/o ANNH) = **B 49.2% / H 46.3% / O 4.6%**
Verifier/normal (w/o ANNH) = **1.06×**
Drafter/normal (w/o ANNH) = **14.2%**
Drafter split (w ANNH) = **B 81.7% / H 10.3% / O 8.0%**
Verifier/normal (w ANNH) = **1.06×**
Drafter/normal (w ANNH) = **8.6%**
ANNH accuracy = **98.5%**
ANNH head speedup = **7.37×**
ANNH index = **8016 clusters; top-k 100**